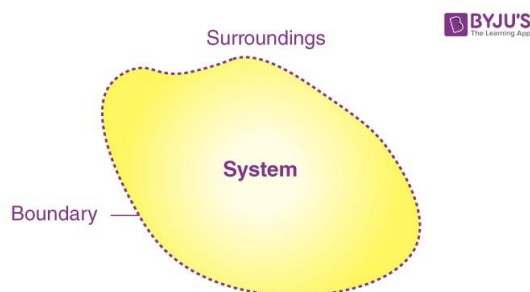


Thermodynamics

Thermodynamics is a branch of physics that deals with the study of energy, heat, work, and the relationships between these quantities. It is a fundamental theory that governs the behavior of matter and energy in physical systems.

Terms in thermodynamics



System: The specific part of the universe under consideration, often separated from its surroundings by a boundary.

Surroundings: Everything outside the system.

State Variables: Properties that define the state of a system and are independent of the path taken to reach that state. Examples include temperature, pressure, and volume.

Process: A change in the state of a system, often represented on a thermodynamic diagram.

Equilibrium: A state in which a system's properties (such as temperature and pressure) do not change with time.

Internal Energy (U): The sum of the kinetic and potential energies of the particles within a system.

Enthalpy (H): The internal energy of a system plus the product of its pressure and volume.

Temperature (T): A measure of the average kinetic energy of particles in a system.

Pressure (P): Force per unit area.

Volume (V): The amount of space occupied by a system.

Entropy (S): A measure of the disorder or randomness in a system.

Heat (Q): Energy transfer between a system and its surroundings due to a temperature difference.

Work (W): Energy transfer due to a force acting through a distance.

First Law of Thermodynamics: The law of energy conservation, stating that energy cannot be created or destroyed, only transferred or converted from one form to another.

Second Law of Thermodynamics: Processes tend to increase the overall entropy of a system and its surroundings.

Third Law of Thermodynamics: As the temperature of a system approaches absolute zero, the entropy of the system approaches a minimum value.

Carnot Cycle: An idealized thermodynamic cycle representing the most efficient possible heat engine.

Adiabatic Process: A process in which no heat is transferred to or from the system ($Q = 0$).

Isothermal Process: A process that occurs at constant temperature.

Heat Capacity (C): The amount of heat needed to change the temperature of a system by 1 degree.

Thermodynamics Process

Isobaric process → pressure remains constant $P_1 = P_2$

Isochoric process → volume remains constant $V_1 = V_2$

Isothermal process → temp remains constant $T_1 = T_2$

Adiabatic process → no heat exchange ($Q=0$) $P, V, T = C$

Reversible process → no energy lost

All **natural** processes are irreversible

an **irreversible** process includes dissipation of energy

All **reversible** processes are quasi static, but not all quasi static processes are **reversible**.

Ex. When there is a small leak

Basic Terms

Thermo- Heat- study of heat, cause, effect, heat flow

Dynamics- Movement- radiation, processes, statistical

Direction of heat flow: Always from hot to cold

Thermodynamic equilibrium: Objects at same temp.

Temperature: measure of relative hotness and coldness

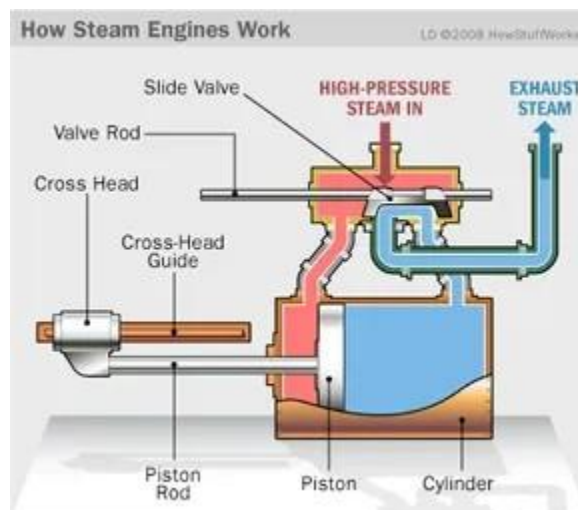
System: is separated from its surroundings

Open system: can exchange mass and energy

Closed system: can ONLY exchange energy

Isolated system: nothing can be exchanged

Steam Engine



A steam engine is a heat engine that converts the energy stored in steam into mechanical work. It operates on the principles of thermodynamics, utilizing the expansion of steam to generate power. The basic components of a steam engine include:

Boiler: This is where water is heated to produce steam. The heat is typically supplied by burning fossil fuels or other sources.

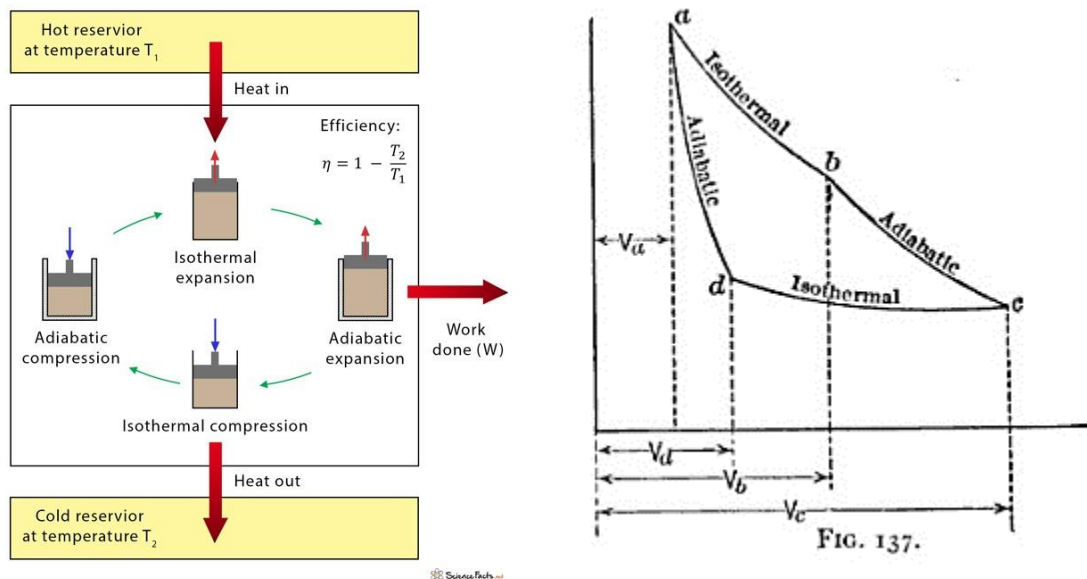
Steam Turbine or Piston: The steam is directed onto a turbine or a piston. In a turbine, the high-speed rotation is converted into mechanical work. In a piston engine, the steam's expansion pushes a piston back and forth, and this linear motion is converted into rotational motion.

Condenser: After the steam has done its work, it is condensed back into water in the condenser. This allows the cycle to repeat.

Generator: The mechanical work produced by the turbine or piston is often used to turn a generator, converting the mechanical energy into electrical energy.

Carnot Cycle:

Carnot Engine Diagram



The Carnot cycle is an idealized thermodynamic cycle that provides an upper limit on the efficiency of heat engines. It consists of four reversible processes: isentropic compression, isothermal heat addition, isentropic expansion, and isothermal heat rejection. However, practical engines do not achieve the Carnot efficiency due to irreversibilities and frictional losses.

Isothermal Expansion (A to B): The working substance (usually a gas) is in contact with a high-temperature reservoir (heat source) at a constant high temperature, T_h . During this process, the gas expands isothermally, meaning its temperature remains constant. The heat Q_h is absorbed from the reservoir, and the gas does work on the surroundings.

Adiabatic Expansion (B to C): The working substance continues to expand adiabatically, meaning without heat exchange. This expansion results in a decrease in temperature and a decrease in pressure. The gas does work on the surroundings, and no heat is added or removed during this process.

Isothermal Compression (C to D): The working substance is in contact with a low-temperature reservoir (heat sink) at a constant low temperature, T_c . During this process, the gas is compressed isothermally, and heat Q_c is rejected to the reservoir.

Adiabatic Compression (D to A): The working substance undergoes adiabatic compression, causing an increase in temperature and pressure. The gas does work on the surroundings, and no heat is added or removed during this process.

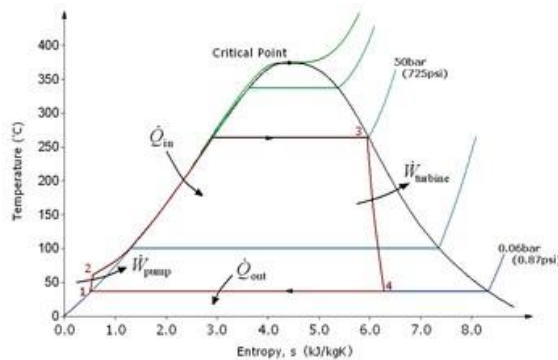
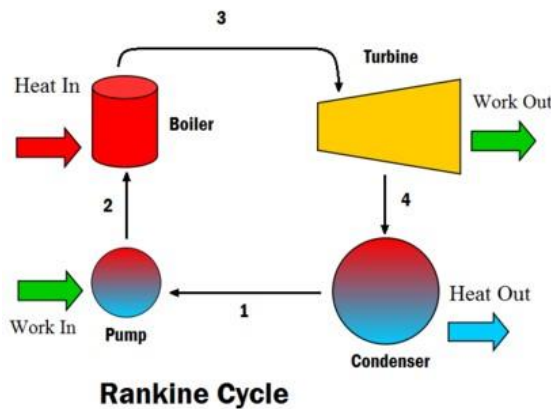
The efficiency (η) of a Carnot engine is given by the ratio of the work done by the engine (W) to the heat absorbed from the high-temperature reservoir (Q_h):

$$\eta = 1 - \frac{T_c}{T_h}$$

where T_c is the absolute temperature of the low-temperature reservoir, and T_h is the absolute temperature of the high-temperature reservoir. The Carnot efficiency is the maximum possible efficiency for a heat engine operating between two given temperature levels. It establishes an upper limit that no real engine can surpass due to the inevitability of irreversible processes and entropy production in actual systems.



Rankine Cycle:



The Rankine cycle is a practical thermodynamic cycle often used in steam power plants. It consists of four processes:

Isentropic Compression: Steam is compressed to a high-pressure liquid.

Isobaric Heat Addition: The high-pressure liquid is heated at constant pressure in the boiler, converting it into high-pressure steam.

Isentropic Expansion: The high-pressure steam expands through a turbine, producing mechanical work.

Isobaric Heat Rejection: The low-pressure steam is condensed back into a liquid in the condenser at constant pressure.

Efficiency:

The efficiency of a heat engine is given by the ratio of the useful work output to the heat input. For the Rankine cycle, the efficiency is given by:

$$\text{Efficiency} = \frac{\text{Work Output}}{\text{Heat Input}}$$

The Carnot efficiency, which is the maximum possible efficiency for a heat engine operating between two temperature reservoirs, is given by:

$$\text{Carnot Efficiency} = 1 - \frac{T_c}{T_h}$$

where T_c is the absolute temperature of the cold reservoir and T_h is the absolute temperature of the hot reservoir.